

Recall

3

$E = \dots$

1. Coordinate functionals in a finite dim NLS
2. Norm defined using the coordinate fnals $\|\cdot\|_E$
3. Any finite dim NLS with $\|\cdot\|_E$ norm is a Banach space
4. Finite dim subspace of NLS is closed.
5. Any norms in a F.D space are equivalent
6. Every FD space is a B.S wrt any norm.

$(C[a, b], \|\cdot\|_\infty)$ is a Banach space.

$(C[a, b], \|\cdot\|_p)$ is a $1 \leq p < \infty$.

is NOT a Banach space.

$(L^p[a, b] - \|\cdot\|_p)$ is a Banach.

Qn: Infinite dimensional space can it
be Banach space wrt one norm
& NOT a Banach space wrt
Another norm.

No.

Results.

1. A Banach space cannot have
a denumerable Hamel basis.
2. If a linear space X has
a denumerable basis then it
cannot be Banach wrt any

a norm.

3. If a l.s. is a B.S. wnt
a norm then it cannot have
a denumerable H.B.

Baire Category theorem.

Background.

(Everywhere) dense

$A \subseteq X$ is everywhere dense if
 $\forall x \in X$ and $\epsilon > 0 \exists$ a pt $a \in A$

$$\exists d(x, a) < \epsilon$$



$$B(x, r) \quad \forall r > 0 \quad \forall x \in X$$

$$B(x, r) \cap A \neq \emptyset$$



$$\text{Let } x \in X \quad \exists a_n \in A \text{ s.t.}$$

$$\text{② } \underline{a_n \rightarrow x \text{ as } n \rightarrow \infty.}$$

NOT dense. $A \subseteq X.$

$$\underline{\exists x \in X} \quad \text{and} \quad \text{② } \exists r > 0$$

$$\text{s.t. } B(x, r) \cap A = \emptyset$$

$$\Rightarrow \overline{A} \neq X.$$

NOWHERE Dense

$$\underline{\forall x \in X} \quad \forall B(x, r)$$

$$\exists B(y, r') \subseteq B(x, r)$$

$$\Rightarrow B(y, r') \cap A = \emptyset.$$

Ex.



$A \subseteq X$ is nowhere dense
if $\underline{(\overline{A})}^\circ = \emptyset.$

All nowhere dense is not dense.

$$(0, 1) \subseteq \mathbb{R}.$$

Ex.

- Finite set
- Finite union of Finite ~~do~~ set.
- X Countable union of finite sets.

Not dense
in $\mathbb{N}$

dense
 $\mathbb{Q}$.

1st Category set

$S \subseteq X$ is of
1st category if
it can be written
as a countable
union of nowhere
dense set.

2nd category

If not
then
2nd
category.

Baire Category theorem

A complete metric space cannot
be of first category.

ie a complete NLS cannot be written as a countable union of Nowhere dense set.

Proof:

Suppose X is a NLS with denumerable Hamel basis

say $E = \{u_1, u_2, \dots\}$

Let $X_n = \text{span} \{u_1, u_2, \dots, u_n\}$.

Since X_n is finite dim. so

X_n is closed.

- Any proper subspace of a NLS

cannot have any int point.

$$A \subsetneq X \quad \text{say} \quad \frac{y \in \text{int}(A) = A^\circ}{\exists r > 0 \quad \text{s.t.} \quad \underline{B(x, r) \subseteq A}}.$$

$$\text{Let } \underline{y \in X}$$

$$z = x + \frac{r}{2} \frac{y}{\|y\|}$$

$$\|z - x\| = \frac{r}{2} < r$$

$$\Rightarrow z \in B(x, r) \subseteq A$$

$$\therefore \underline{x \in A}, \underline{z \in A}, \quad A \text{ proper subspace.}$$

$$\underline{y = \left(\frac{2\|y\|}{r} \right) (z - x) \in A.}$$

$$\therefore \text{int}(A) = \emptyset$$

$$\bullet \quad X_n \rightarrow \text{closed} \quad \& \quad X_n \subseteq X$$

$$\overline{X_n}^\circ = X_n^\circ = \phi$$

$\therefore X_n$'s are nowhere dense sets

$\&$ since $X = \bigcup_n X_n \Rightarrow X$ is not complete
(By Baire category theorem).

Consequences.

1. $(\mathcal{P}[a, b], \|\cdot\|_\infty)$ not a Banach space.

Denumerable basis

$$E = \{u_1, \dots\}$$

$$u_j(t) = t^{j-1}$$

2. C_{00} has a denumerable Hamel basis

$$\{e_1, e_2, \dots\}$$

$$e_j = (0, 0, \dots, \underset{j\text{th}}{1}, 0, \dots, 0).$$

If $x \in C_{00}$

$$\begin{aligned} x &= (x(1), x(2), \dots) \\ &= x(1)e_1 + x(2)e_2 + \dots \end{aligned}$$

C_{00} is NOT a Banach space.

$$x \in l^p.$$

$$x = (x(1), x(2), \dots)$$

$$\sum |x(j)|^p < \infty$$

$\{e_1, e_2, \dots\}$ is not a Hamel basis for l^p .

$$p=1$$

$$x = (1, \frac{1}{2}, \frac{1}{2^2}, \dots) \in l^1$$

$$\sum \frac{1}{2^n} < \infty$$

$x \notin \text{l.c. of finitely many}$
 $\{e_1, e_2, \dots\}$.

If X is a Banach space

i) Either X is finite dim.

(2.) X has a basis consisting of uncountable number of elements.

Schauder Basis.

A seq $\{u_n\}$ in a NLS is said to be a Schauder Basis of X if

$\forall x \in X \quad \exists! (\alpha_n), \alpha_n \in \mathbb{K}.$

s.t $x = \sum_{j=1}^{\infty} \alpha_j u_j$ ie

$$S_N = \sum_{j=1}^N \alpha_j u_j \Rightarrow \|S_N - x\| \rightarrow 0 \text{ as } N \rightarrow \infty.$$

$$\left[X = \text{span} \{u_1, \dots, u_n\} \right. \\ \left. \text{span} \{u_1, \dots, u_n, \dots\} = X \right]$$

Remark:

1. Schauder basis if exists then it is L.I.

$$\sum_{i=1}^n \beta_i u_i = 0$$

Let $\beta_j = 0$, $j > n$

Then

$$\sum_{j=1}^{\infty} \beta_j u_j = 0 = \sum_{j=1}^{\infty} 0 \cdot u_j$$

$$\Rightarrow \beta_j = 0 \quad \forall j \in \mathbb{N}.$$

$\{u_1, u_2, \dots\}$ are lin imp.

(2)

Then

$$x \in X$$

$$x = \sum_{j=1}^{\infty} \alpha_j u_j$$

$$= \lim_{n \rightarrow \infty}$$

$$\sum_{j=1}^n \alpha_j u_j$$

$$= \lim_{n \rightarrow \infty} y_n$$

$$y_n = \sum_{j=1}^n \alpha_j u_j$$

$\uparrow$
 $\text{Span} \{u_1, \dots, u_n\}$

$$\Rightarrow x \in \text{Span} \{u_1, \dots, u_n, \dots\}$$

$\therefore \text{span}\{u_1, \dots, u_n, \dots\}$ is dense in X .

③ Let X possess a Hamel basis say

$$E = \{u_1, \dots\}.$$

Then E is a Schauder basis.

Ex.

1.) $(C_0, \|\cdot\|_\infty)$

$E = \{e_1, \dots, e_n, \dots\} \rightarrow$ Hamel basis
 $\Downarrow$

Schauder basis.

2. $(P[a, b], \|\cdot\|_\infty)$

$$u_j(t) = t^{j-1}$$

$$3. \quad (l^p, \|\cdot\|_p)$$

$$E = \{e_1, e_2, \dots\} \quad \left\{ \begin{array}{l} x \in X \\ x = (x(1), x(2), \dots) \end{array} \right.$$

$$\sum_{j=1}^n x(j) e_j = (x(1), x(2), x(3), \dots, x(n), 0, \dots)$$

$$\|x - \sum_{j=1}^n x(j) e_j\|_p$$

$$= \| (x(1), x(2), x(3), \dots, x(n), x(n+1), \dots) - (x(1), x(2), \dots, x(n), 0, \dots, 0) \|_p$$

$$= \| (0, 0, \dots, 0, x(n+1), \dots) \|_p$$

$$= \sum_{j=n+1}^{\infty} |x(j)|^p < \epsilon \quad \forall n \geq N.$$

$$\therefore \left\| x - \sum_{j=1}^n x(j) e_j \right\|_p \rightarrow 0 \text{ as } n \rightarrow \infty$$

Uniqueness ??

$\{e_1, e_2, \dots, e_n\}$ $\rightarrow$ Schauder basis.

$(C[0,1], \|\cdot\|_{\infty})$ $\begin{cases} \text{Faber Schauder system} \\ \text{Franklin system.} \end{cases}$

Separability

$$A \subseteq X$$

, $A \xrightarrow{\text{dense}}$
 $\searrow$ countable.

Ex. • $X = \mathbb{R}$.

$$A = \mathbb{Q}$$

• $X = \mathbb{C}$ $A = \mathbb{Q} + i\mathbb{Q}$

Ex. • Every subspace of a sep. m.s.
is separable.

• $A \subseteq X$
dense, A is separable
 $\Rightarrow X$ is separable.

Theo. Every finite dimensional NLS is separable.

Proof. Let X be a f.d NLS.
 $E = \{u_1, \dots, u_n\}$ be a basis for X .

For each $x \in X$

$$x = \sum_{j=1}^n \alpha_j u_j$$

$$\alpha_j \in \mathbb{K}.$$

$$\mathbb{Q}_K = \begin{cases} \mathbb{Q} & , \quad \mathbb{K} = \mathbb{R} \\ \mathbb{Q} + i\mathbb{Q} & , \quad \mathbb{K} = \mathbb{C} \end{cases}$$

$$D = \left\{ \sum_{j=1}^n \beta_j u_j \mid \beta_j \in \mathbb{Q}_k \right\}.$$

- D is countable.
- D dense (?)

$$x = \sum_{j=1}^n \alpha_j u_j \quad |$$

By denseness of $\mathbb{Q}_k$ for each j , $\exists \beta_j$ s.t

$$|\alpha_j - \beta_j| < \epsilon/2 \quad \forall j = 1, 2, \dots, n$$

$$\begin{aligned} \left\| x - \sum_{j=1}^n \beta_j u_j \right\| &= \left\| \sum_{j=1}^n \alpha_j u_j - \sum_{j=1}^n \beta_j u_j \right\| \\ &\leq \sum_{j=1}^n |\alpha_j - \beta_j| \|u_j\| \end{aligned}$$

$$< \epsilon \left(\sum_{j=1}^n \|g u_j\| \right)$$

$\therefore D$ is dense in X .